This script creates the `chats` index with one primary shard on a single node. Elasticsearch doesn't create the replica of this index on the same node where the primary shard exists. Indeed, there's no point in creating a backup drive in the same location as the main drive. Figure 14.2 shows this (a shard is created, but a replica isn't).

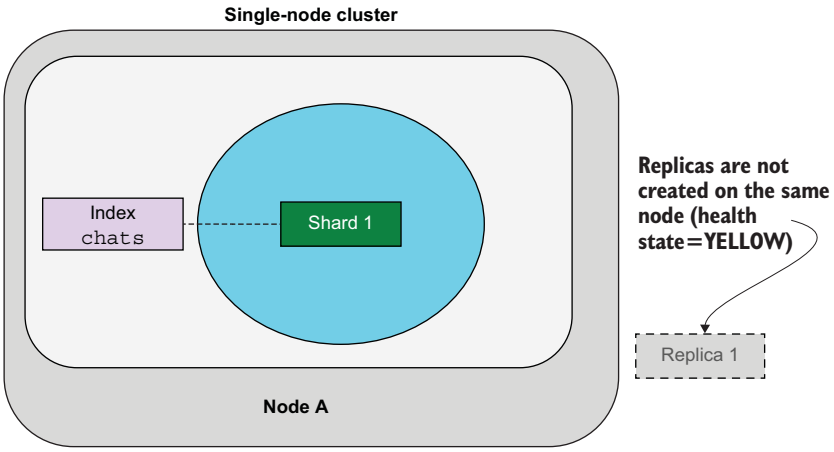


Figure 14.2 Single-node cluster with no replica created

If replicas aren't created, the cluster isn't classified as being in a healthy state. We can use the cluster health API to get a high-level view of the cluster state. The `GET _cluster/health` call fetches the health of the cluster and outputs it in a JSON format that details the unassigned shards, the state of the cluster, the number of nodes and data nodes, and so on. But what do we mean by cluster health?

14.1.2 Cluster health

As we discussed in chapter 3, Elasticsearch uses a simple traffic light system to let us know the state of the cluster: GREEN, RED, and YELLOW. When we first create an index on a single-node server, its health is YELLOW because the replica shards are not yet assigned (unless we purposely set `replicas` to zero on all indexes on this node, which is possible but is an anti-pattern). If needed, refer to section 3.2.5 for a refresher on the shard traffic light health system. Figure 14.3 is repeated from chapter 3 and defines the health of a cluster based on the shard assignments.

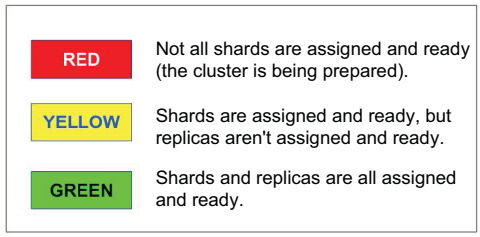


Figure 14.3 Representing the health of a cluster using traffic lights